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Cyberweerbaarheid en Architectuur:
De chaos omarmen

Nick Nieuwenhuis

Cybersecurity Architect

Nedscaper



Waarom hebben we het over cyberweerbaarheid?

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Jaguar Land Rover production severely hit by cyber-attack

Grote verstoringen luchthavens waren gevolg van ransomware-aanval

Cyberrisico's nemen toe: CISO's in Nederland verwachten meer aanvallen

Clinical Diagnostics doet nog onderzoek naar cyberaanval medische onderzoeken

Introductie

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Nedscaper

Microsoft CERTIFIED CYBERSECURITY ARCHITECT EXPERT

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HU LECTORAAT CYBER SECURITY

The Open Group Open Certified TOGAF® Enterprise Architecture Practitioner

Achtergrond

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Cyberrisico’s worden groter

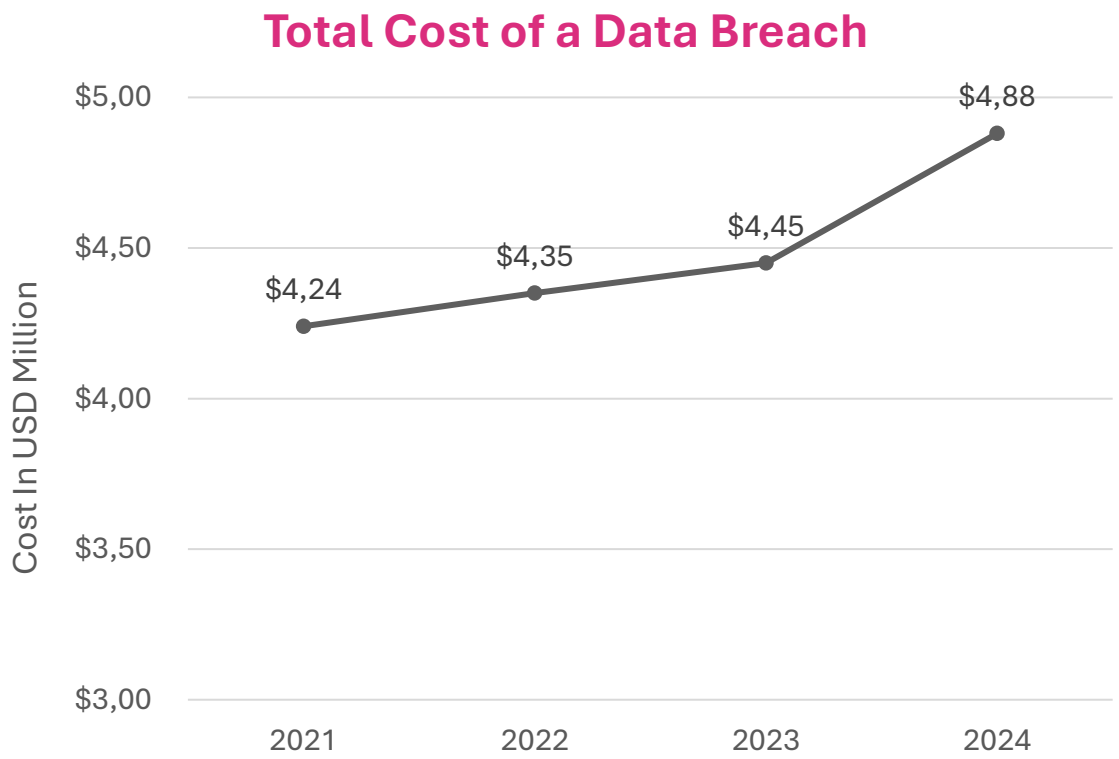
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1) Organisaties worstelen continu met cyberrisico’s

2) De totale kosten gemoeid met een cyberaanval stijgen

3) Groeiende wet- en regelgeving om cyberrisico’s te verkleinen

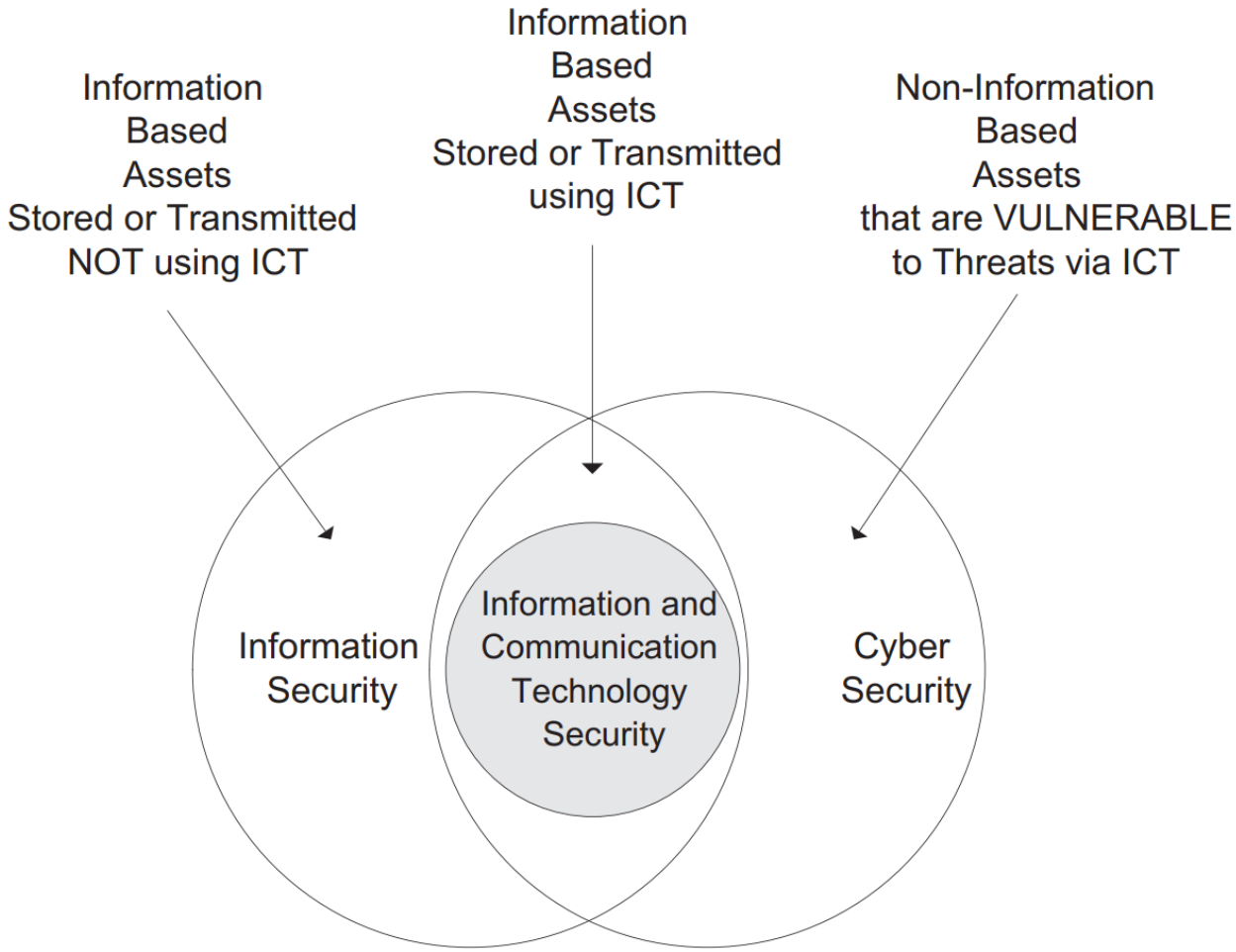
(Cbw, DORA, CRA)



Bron: IBM (2024)

Informatiebeveiliging en Cyberbeveiliging

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Bron: Von Solms & Van Niekerk (2013)

Oorzaken van cyberaanvallen

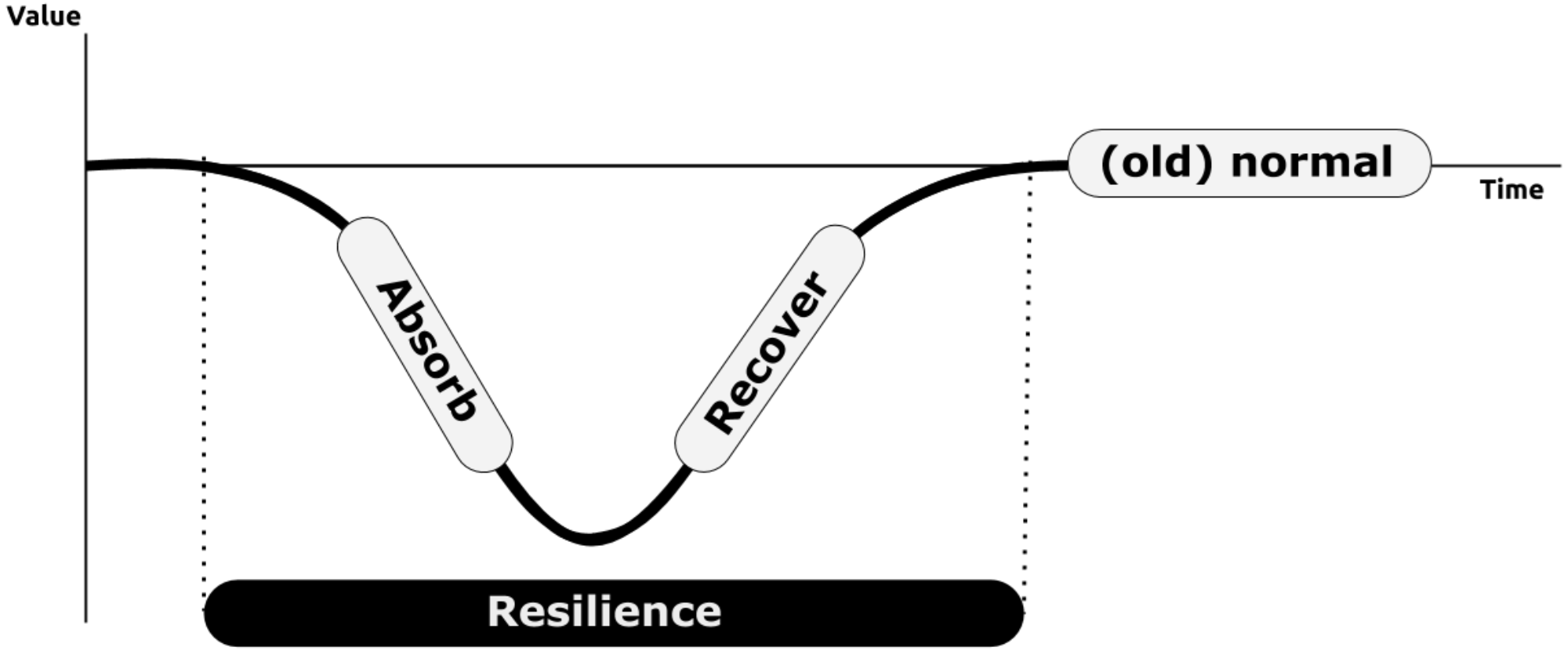
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Actions of People	System & Technology Failures	Failed Internal Processes	External Events
• Mistakes/errors • Fraud • Sabotage • Theft • Inaction	• Software vulnerabilities • Hardware failure • System failure • Bad coding	• Information flow • classification of information • Training & Development • Procurement	• Disasters • Attacks • Espionage • Supply chain risks

Bron: Cebula et al. (2014)

Weerbaarheid / Resilience

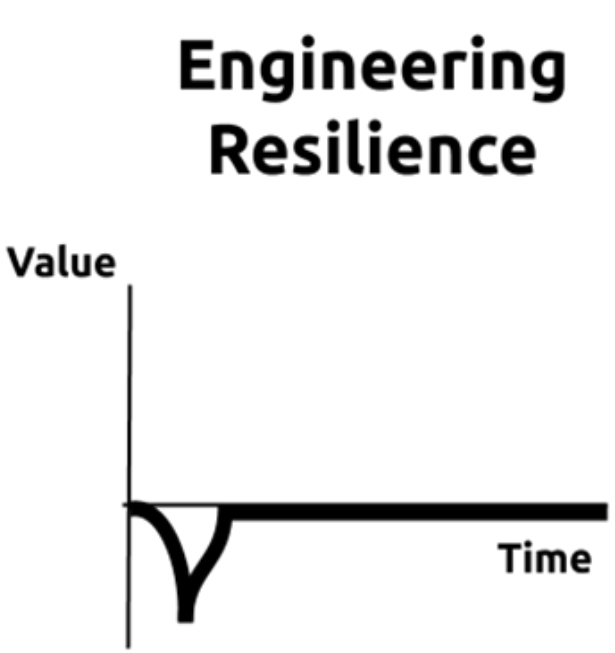
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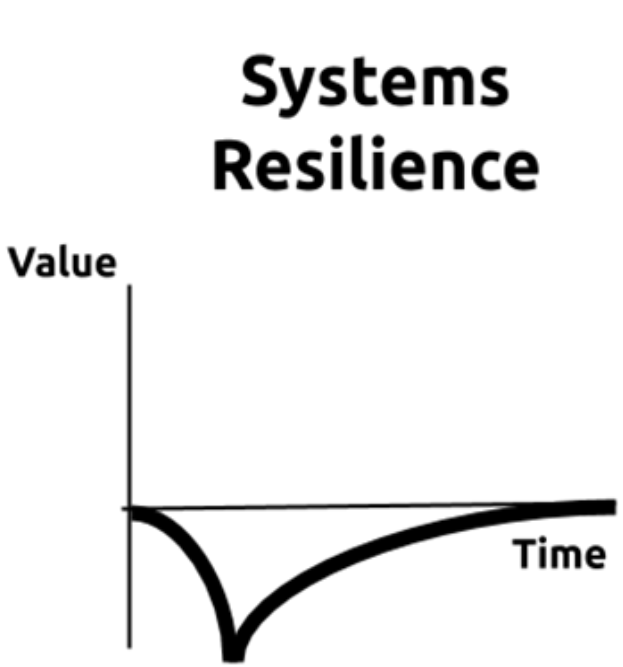


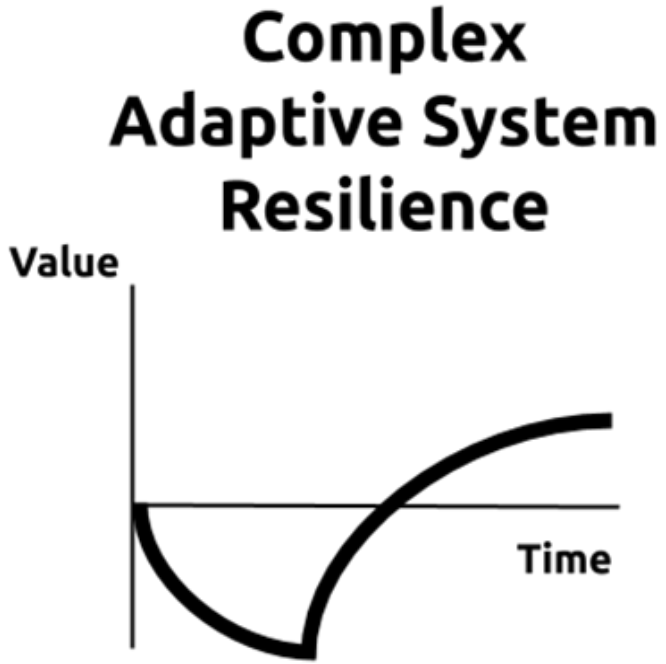
Bron: Botjes et al. (2021)

Drie typen van Resilience

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Bron: Botjes et al. (2021), Martin-Breen & Anderies (2011)

Cyberweerbaarheid

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Identify	• Anticipating and understanding cybersecurity risks allows organizations to proactively prepare for and build resilience against potential adverse events.
Protect	• The implementation of safeguards not only protects against cybersecurity threats but also significantly contributes to an organization's capacity to withstand and recover from disruptions.
Detect	• Timely detection of cybersecurity events enhances an organization's agility in responding quickly and effectively, thereby contributing to overall resilience.
Respond	• The response function is crucial for mitigating the impact of cybersecurity incidents, facilitating a swift return to normal operations and bolstering overall resilience.
Recover	• The recovery function focuses on restoring capabilities after a cybersecurity incident, representing a fundamental element of overall resilience.

Bron: Tzavara & Vassiliadis (2023)

Definities

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Cyber Security	*Securing **information** and **noninformation** assets that are within **cyberspace** or could be affected via cyberspace* (Eling et al., 2021)
Cyber Resilience	*The capacity to **withstand, recover** from and **adapt** to external shocks caused by cyber risks* (Dupont et al., 2023)

Enterprise Architectuur



Wat is EA?

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Enterprise	An enterprise is an intentionally created cooperative of human beings with a certain societal purpose (Dietz et al.. 2013)
Enterprise Architecture	Fundamental concepts or properties of an **enterprise** in its **environment** and the governing principles for the **realization** and **evolution** of the **enterprise** (ISO 42010)

Waarom Enterprise Architectuur?

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Provides a holistic view of the enterprise



Improved business-IT alignment



Improved risk management



Improved asset management

Waarom Enterprise Architectuur?

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"Adopting an EA approach (…) integrates Cyber Security into every aspect of information system design."

Bron: Loft et al. (2022)

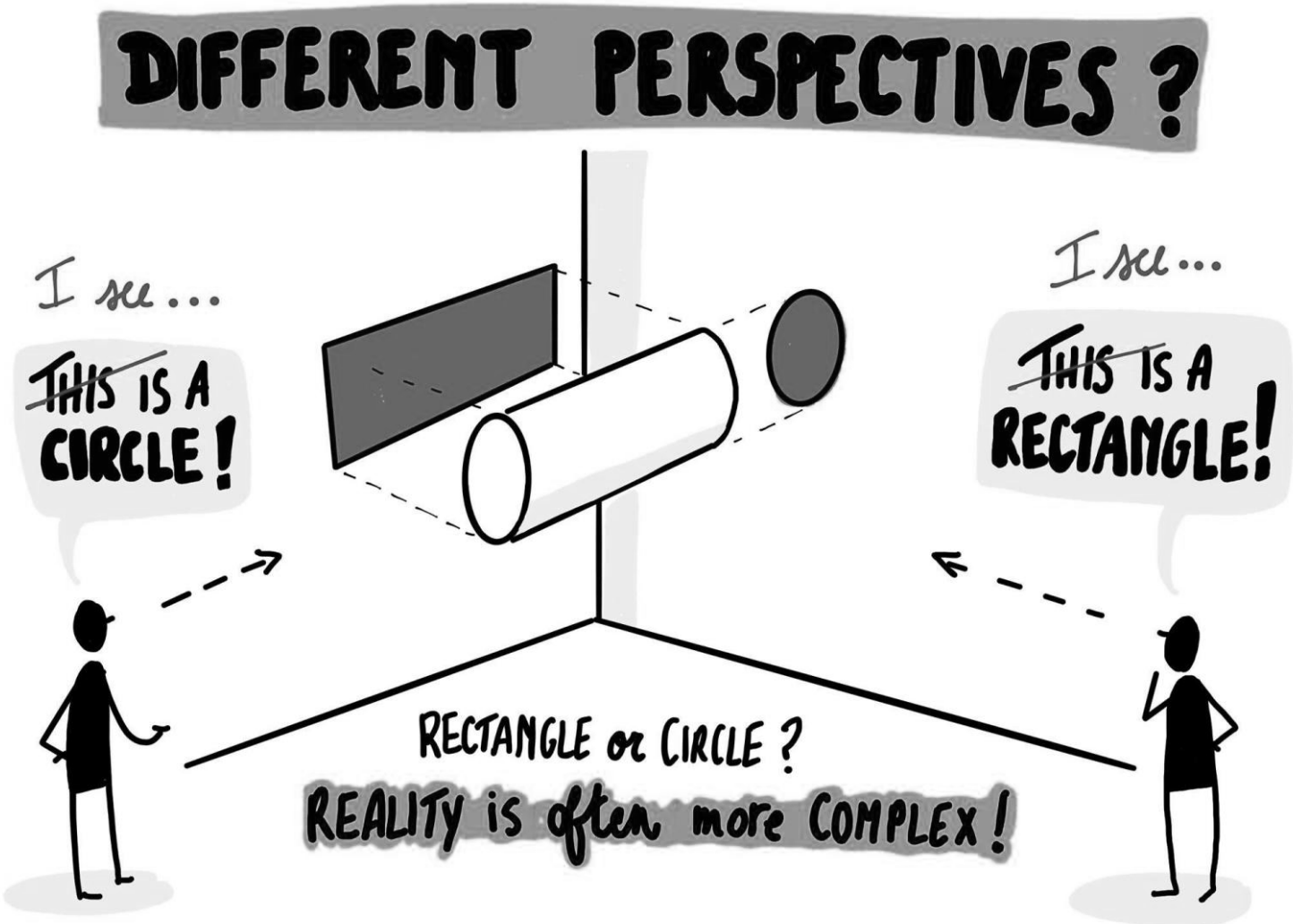
Perceptie & Realiteit

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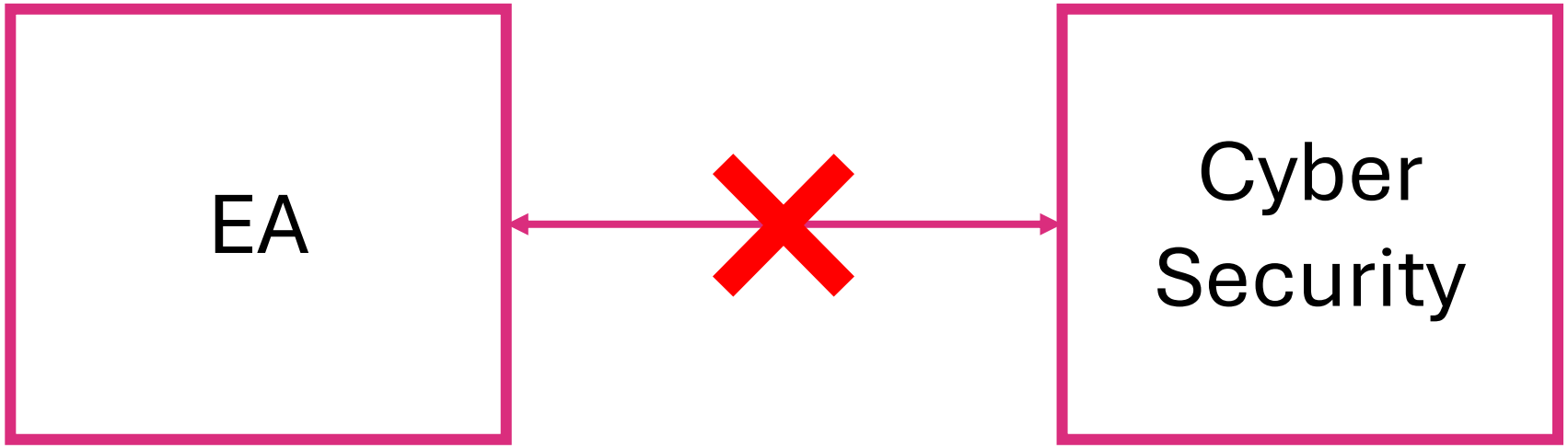
Perceptie & Realiteit

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Het probleem:

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Probleemstelling: “The lack of integration between EA and Cyber Security leaves enterprises at risk.”

Onderzoek

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Hoofdvraag

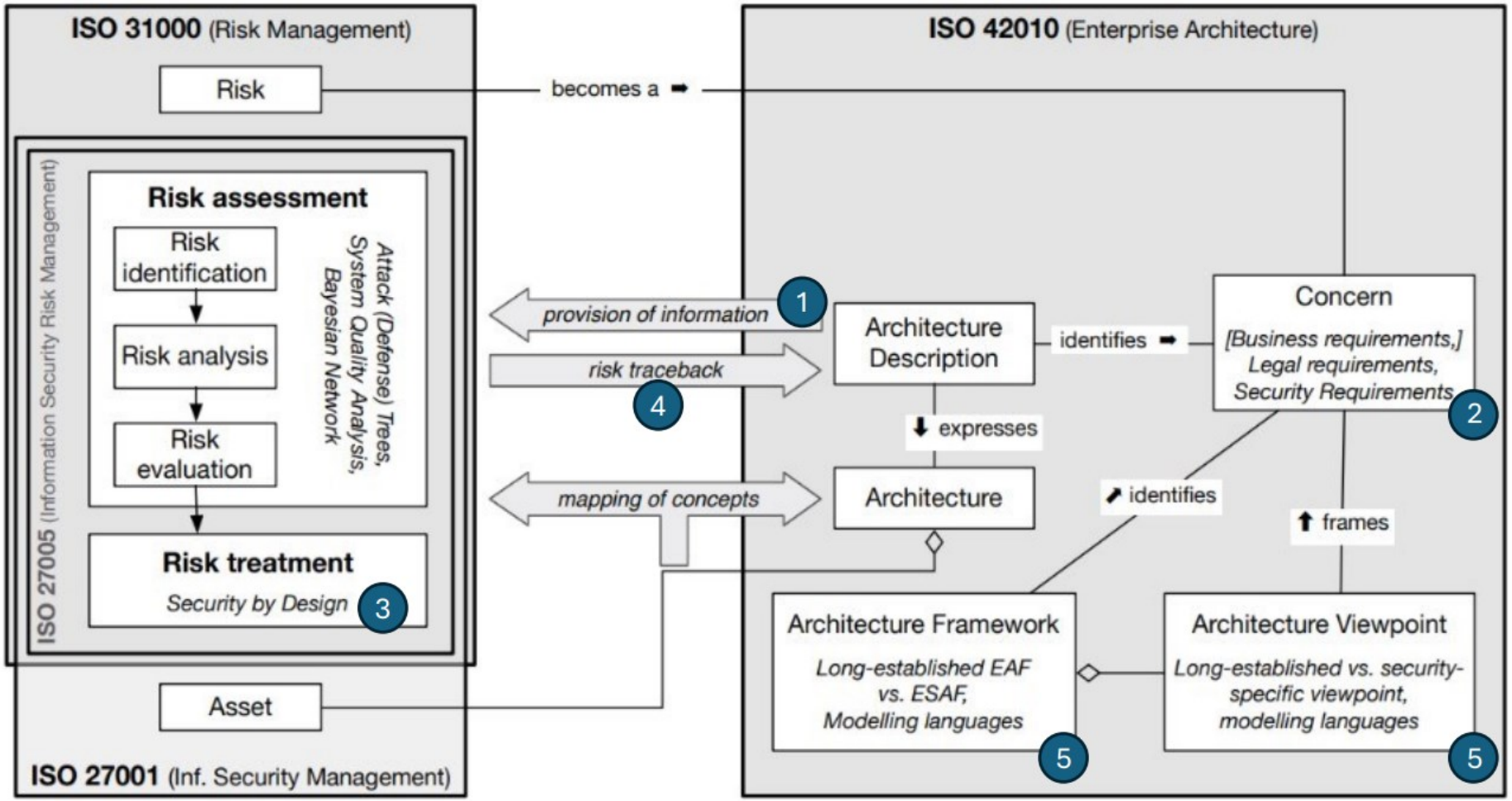
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How can **Cyber Security** and **Enterprise Architecture** be integrated in relation to **Cyber Risk Management** within enterprises?

Strategieën voor het integreren van CS and EA

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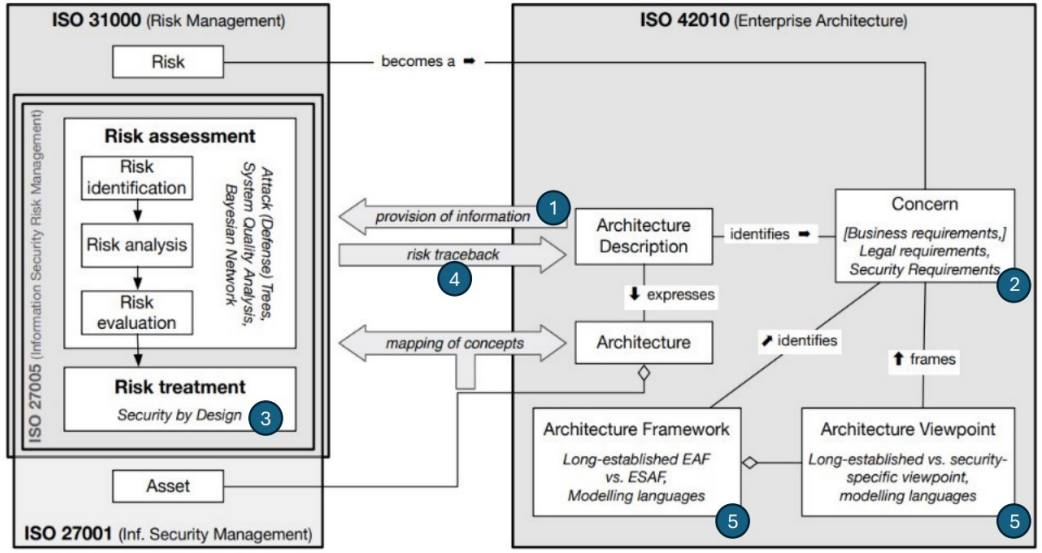


Bron: Diefenbach et al. (2019)

Strategieën voor het integreren van CS and EA

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1. Leveraging EA to provide input for Cyber Risk Assessment
2. Integrating business requirements with security requirements
3. Adopting the ‘Security by Design’ paradigm
4. Mapping identified cyber risks to EA components
5. Integrating Cyber Security into EA frameworks
6. Managing cyber risks by aligning Business & IT activities



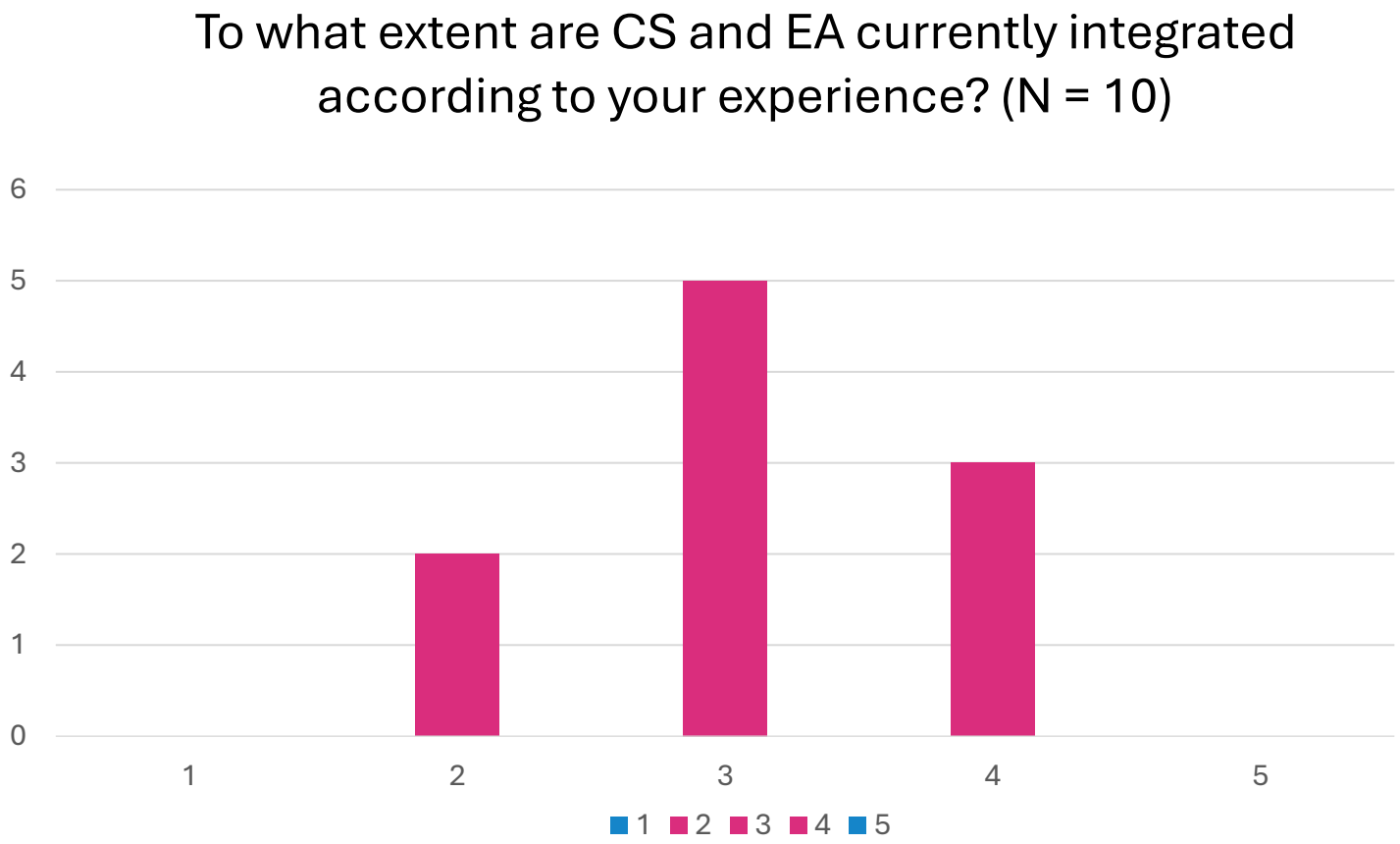
Bron: Diefenbach et al. (2019)

Resultaten

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Huidige integratie van CS en EA

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Average score
3.1

Huidige integratie van CS en EA

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Ranking the six strategies from the LR in order of perceived importance (N = 10)

Priority	Strategy	Score	Spread
1	Integrating Cyber Security in EA frameworks	**29.2**	8%
2	Adopting the ‘Security by Design’ principle	**25.8**	12%
3	Integrating Business Requirements with Security Requirements	**18.4**	26%
4	Leveraging EA to provide input for Cyber Risk Assessment	**15**	18%
5	Mapping identified cyber risks to EA components	**5.8**	8%
6	Managing cyber risks by aligning business & IT activities	**5.8**	12%

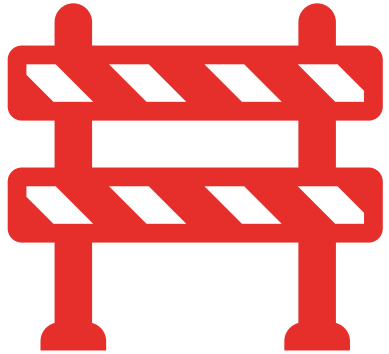
Blockers

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Different Mindsets & Focus

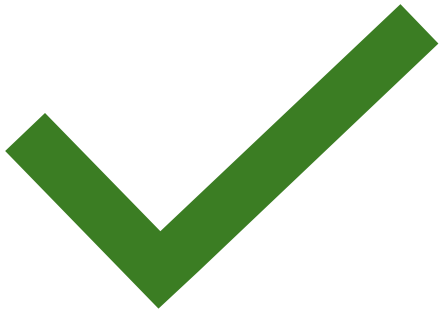
Organizational Misalignment

Skills & Knowledge Gaps



Enablers

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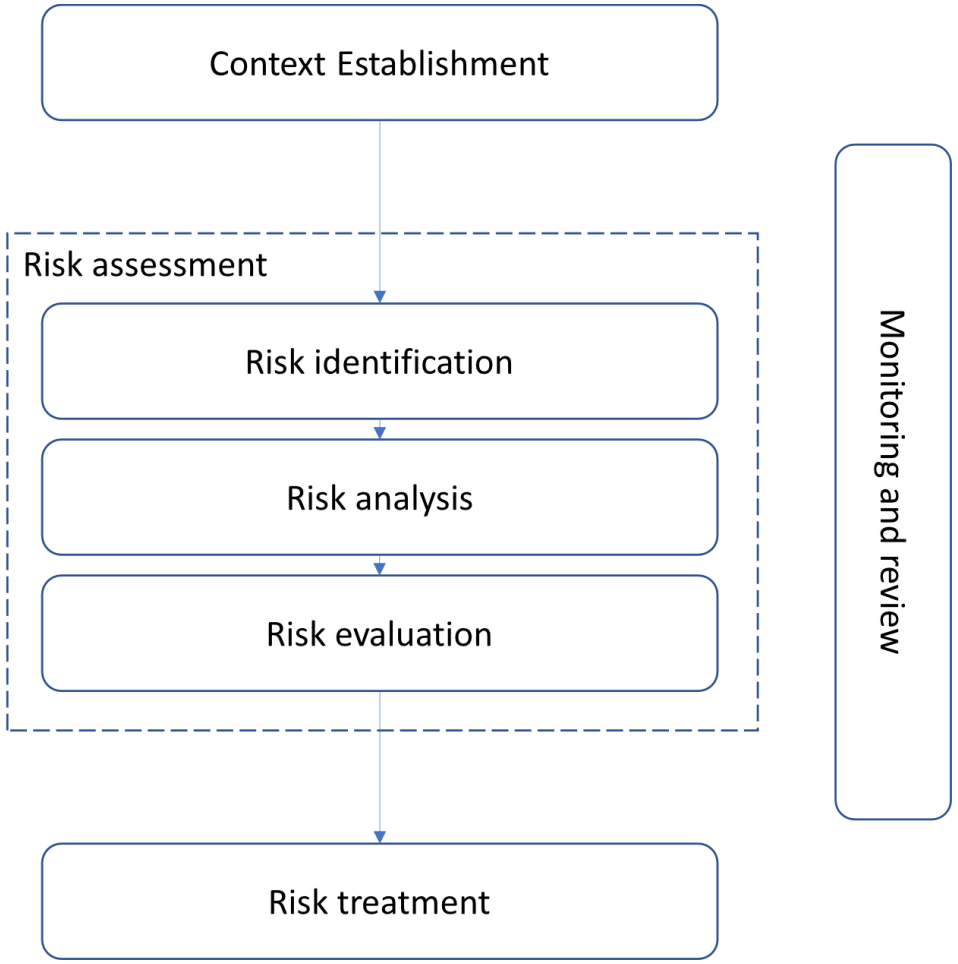
Security Integration into EA Frameworks

Adopting Secure Development Methodologies

Security Awareness in the Business

Impact op cyber risicomanagement

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Context Establishment

EA plays an important role in assessing risks on the process and organizational level

Risk Assessment

CS-EA integration makes Risk Identification and Assessment easier because EA can clarify dependencies and risks involved

Risk Treatment

CS-EA integration realizes faster and more structural solutions for risk mitigation and can align security controls with the architecture

Monitoring and Review

CS-EA integration makes it easier to monitor security controls for their desired effect

Conclusie

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Conclusion

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How can **Cyber Security** and **Enterprise Architecture** be integrated in relation to **Cyber Risk Management** within enterprises?

Conclusie

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By embedding CS into EA Frameworks

Integrating security into EA frameworks ensures security is a primary concern and not an afterthought.

By improving knowledge of CS and EA teams

Improving knowledge is crucial for shared understanding, awareness, and knowledge exchange.

By aligning CS and EA functions in the Organizational Structure

Creating a shared vision, mindset and strategy will enhance collaboration and joint decision making.

By leveraging agile and secure development methodologies

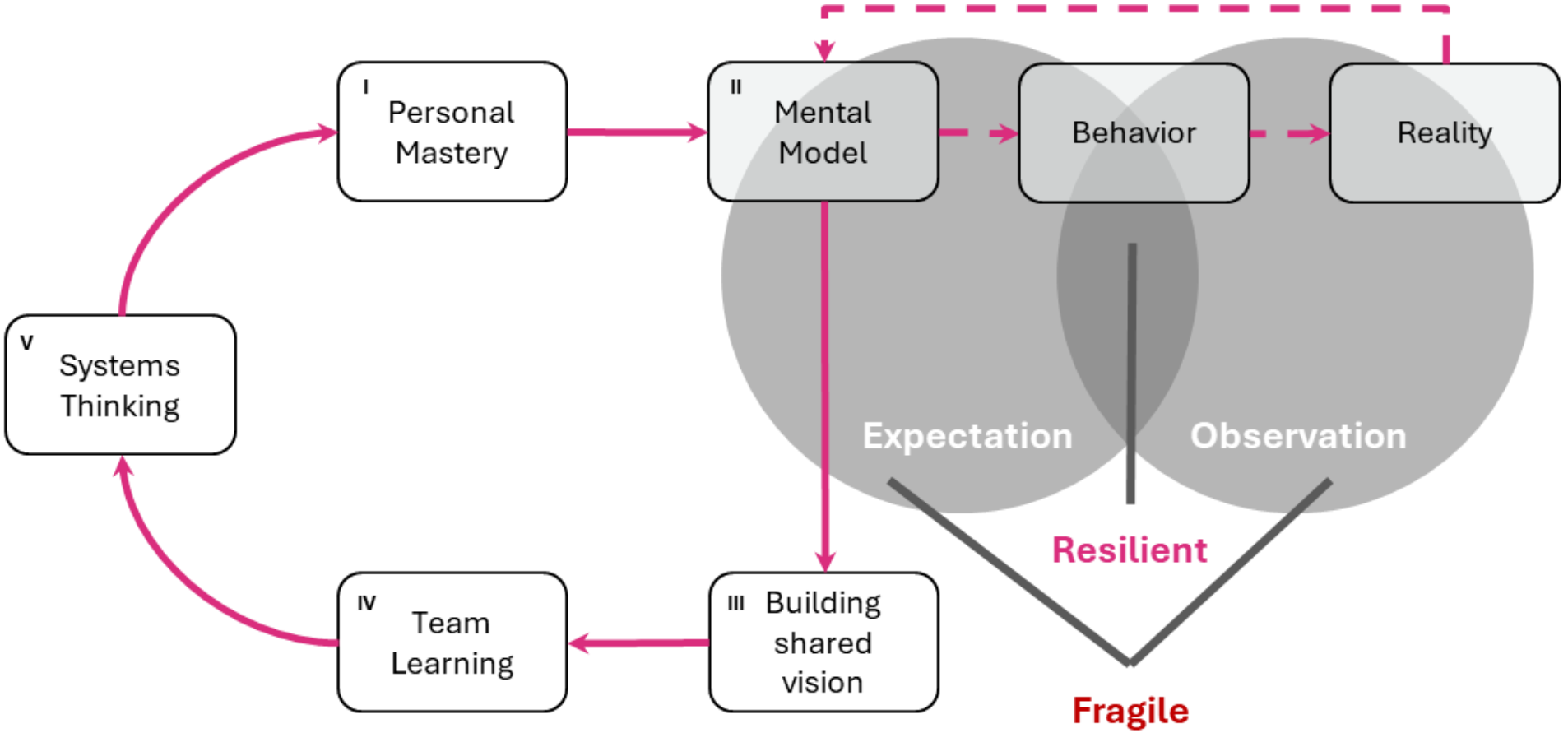
Ensuring enterprises can implement CS measures proactively and holistically.

Wat nu?

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Hoe nu verder?

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Bron: Senge (1990), Botjes (2025)

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Bedankt!

Cyberweerbaarheid en Architectuur:
De chaos omarmen

Nick Nieuwenhuis
30 September 2025